Copilot

Kadane's Algorithm, named after its inventor Jay Kadane, is a dynamic programming technique used to find the maximum sum of a contiguous subarray within a given array of numbers. The goal is to identify the subarray with the largest sum. Here's how it works:

1. Initialize two variables:
   - `max_so_far`: Stores the maximum sum of a contiguous subarray found so far (initially set to negative infinity).
   - `max_ending_here`: Stores the maximum sum contiguous subarray ending at the current index (initially set to zero).
2. Iterate through each element of the array:
   - Update `max_ending_here` by adding the current element.
   - If `max_so_far` is less than `max_ending_here`, update `max_so_far` to `max_ending_here`.
   - If `max_ending_here` becomes negative, reset it to zero.
3. The final value of `max_so_far` represents the maximum sum of a contiguous subarray.

Example: Consider the array `{-2, -3, 4, -1, -2, 1, 5, -3}`:

- For index 0, `max_ending_here` becomes 0 (since -2 < 0) and `max_so_far` remains -2.
- For index 1, `max_ending_here` becomes 0 (since -3 < 0).
- For index 2, `max_ending_here` becomes 4 (since 4 > 0), and `max_so_far` updates to 4.
- Continue this process for the entire array.

The subarray `{4, -1, -2, 1, 5}` has the largest sum of 7 in this example. Kadane's Algorithm is widely used in solving related problems and has applications in various domains, including computer science and finance. 🚀